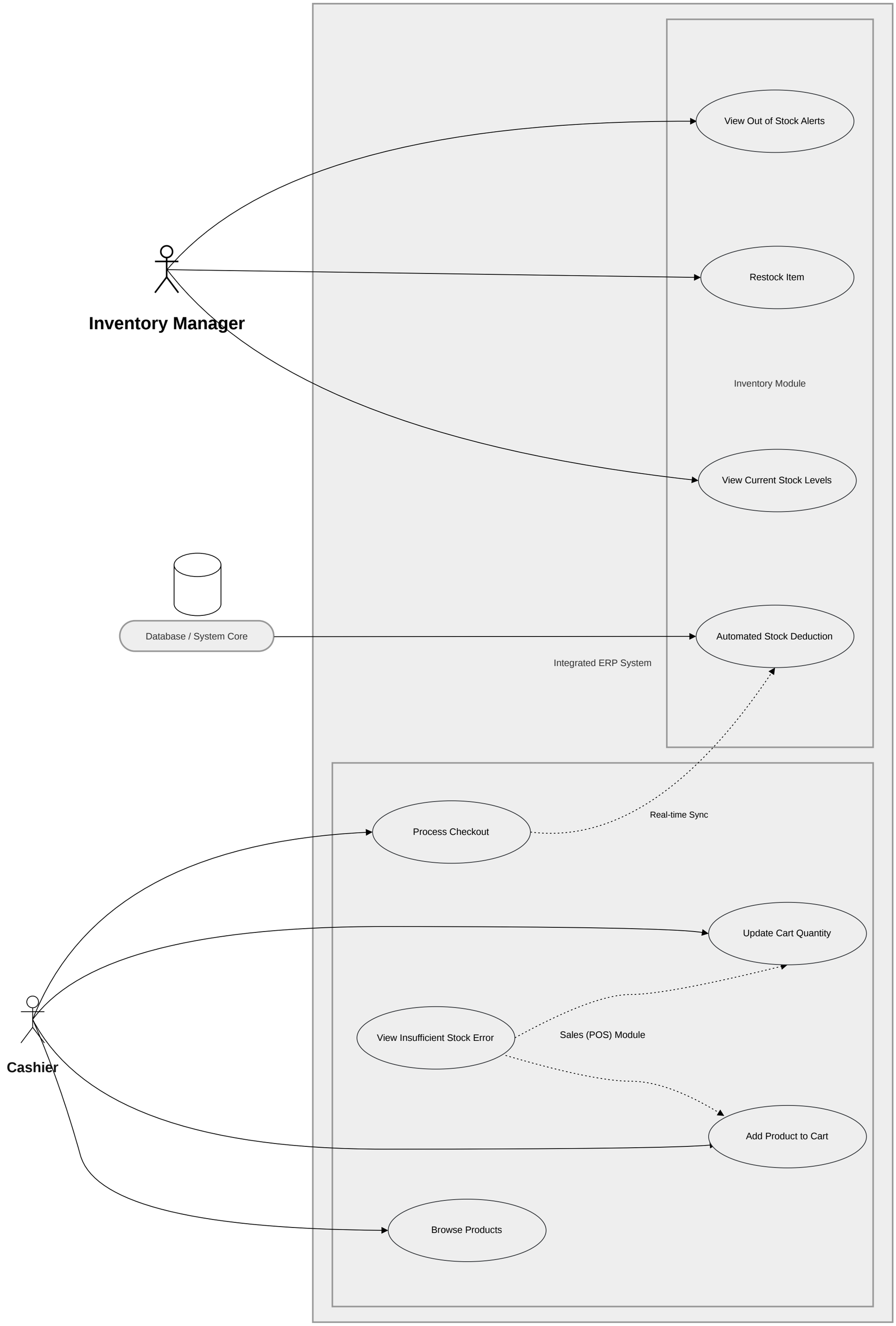
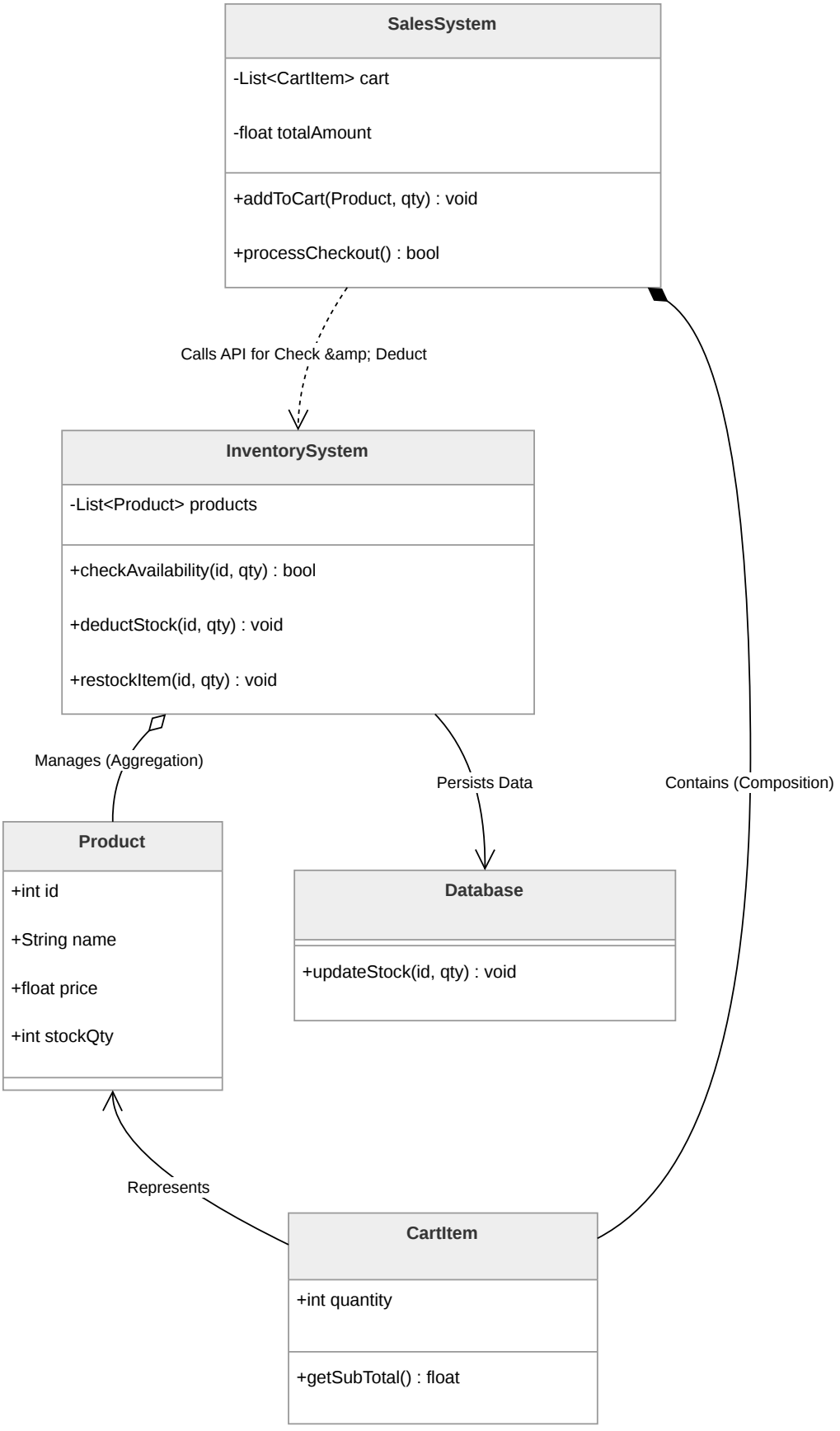
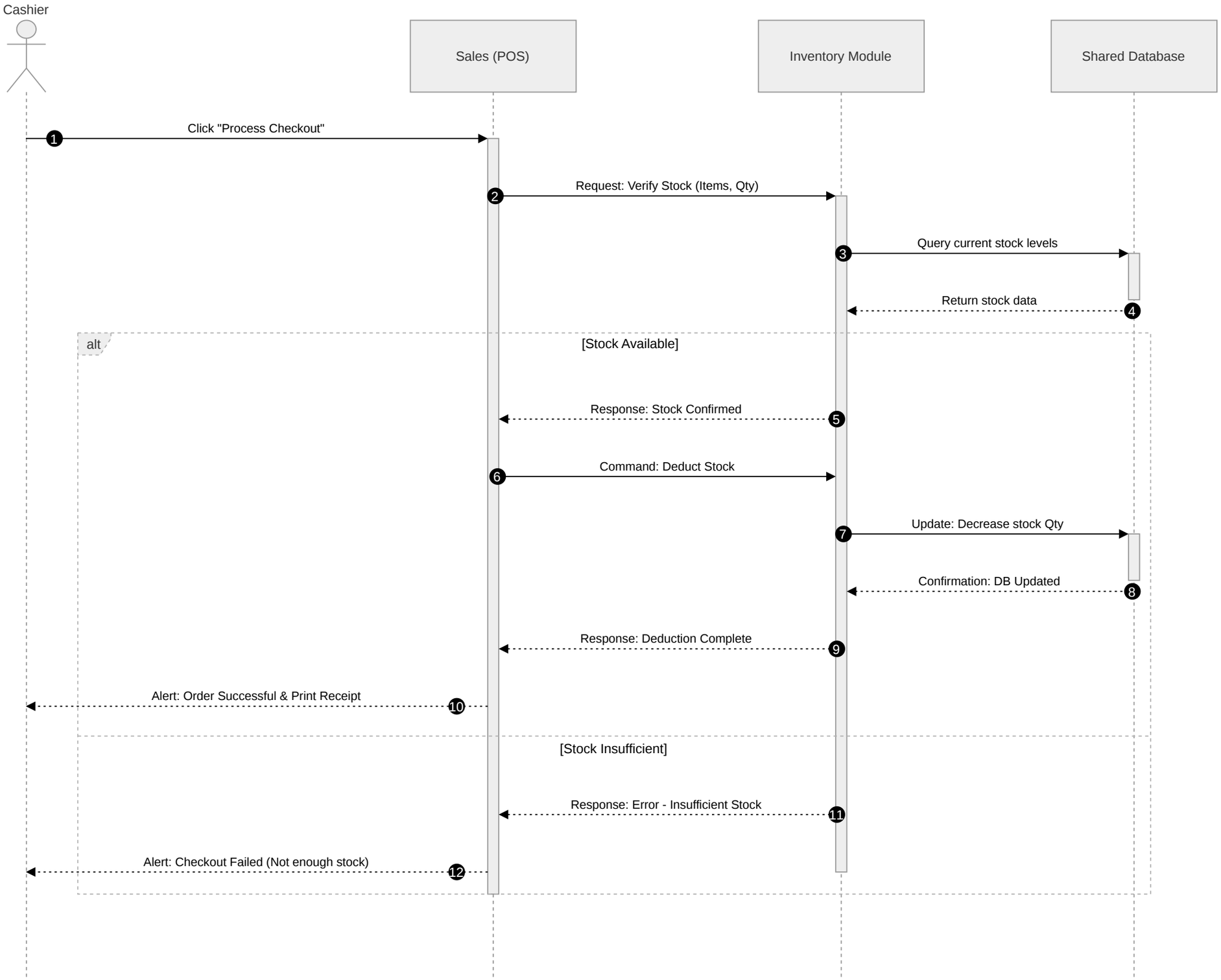
Use Case Diagram



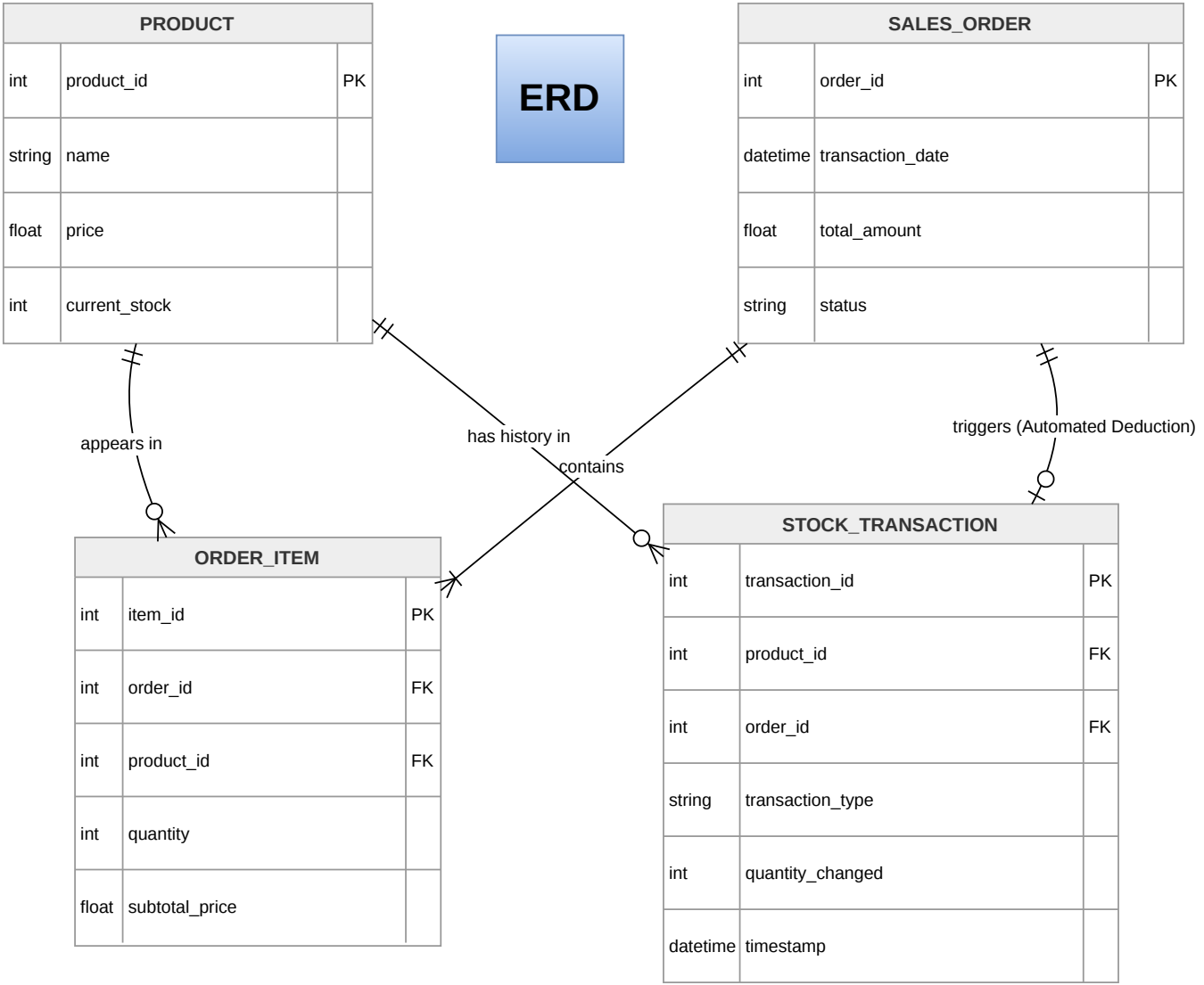
Class Diagram



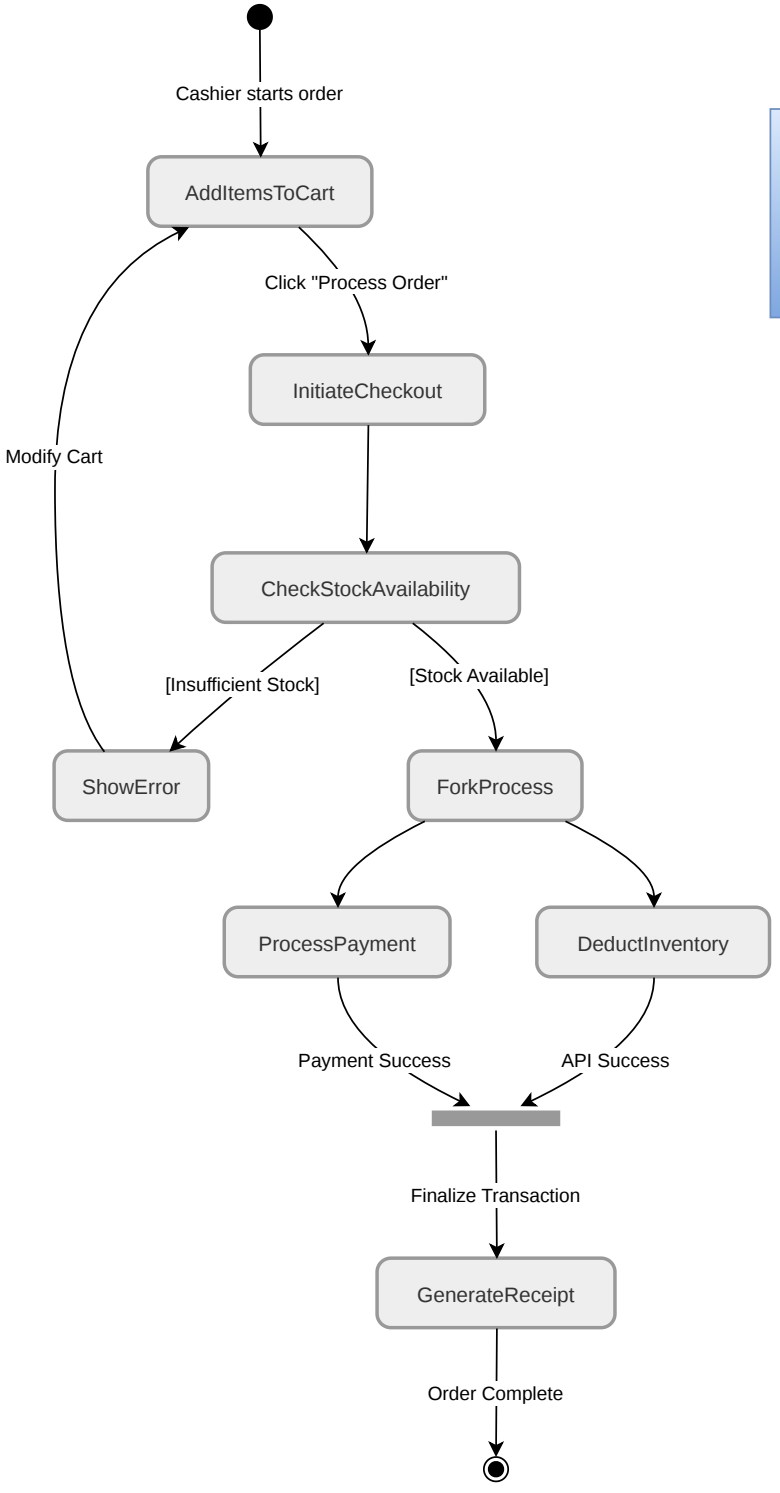


Sequence Diagram

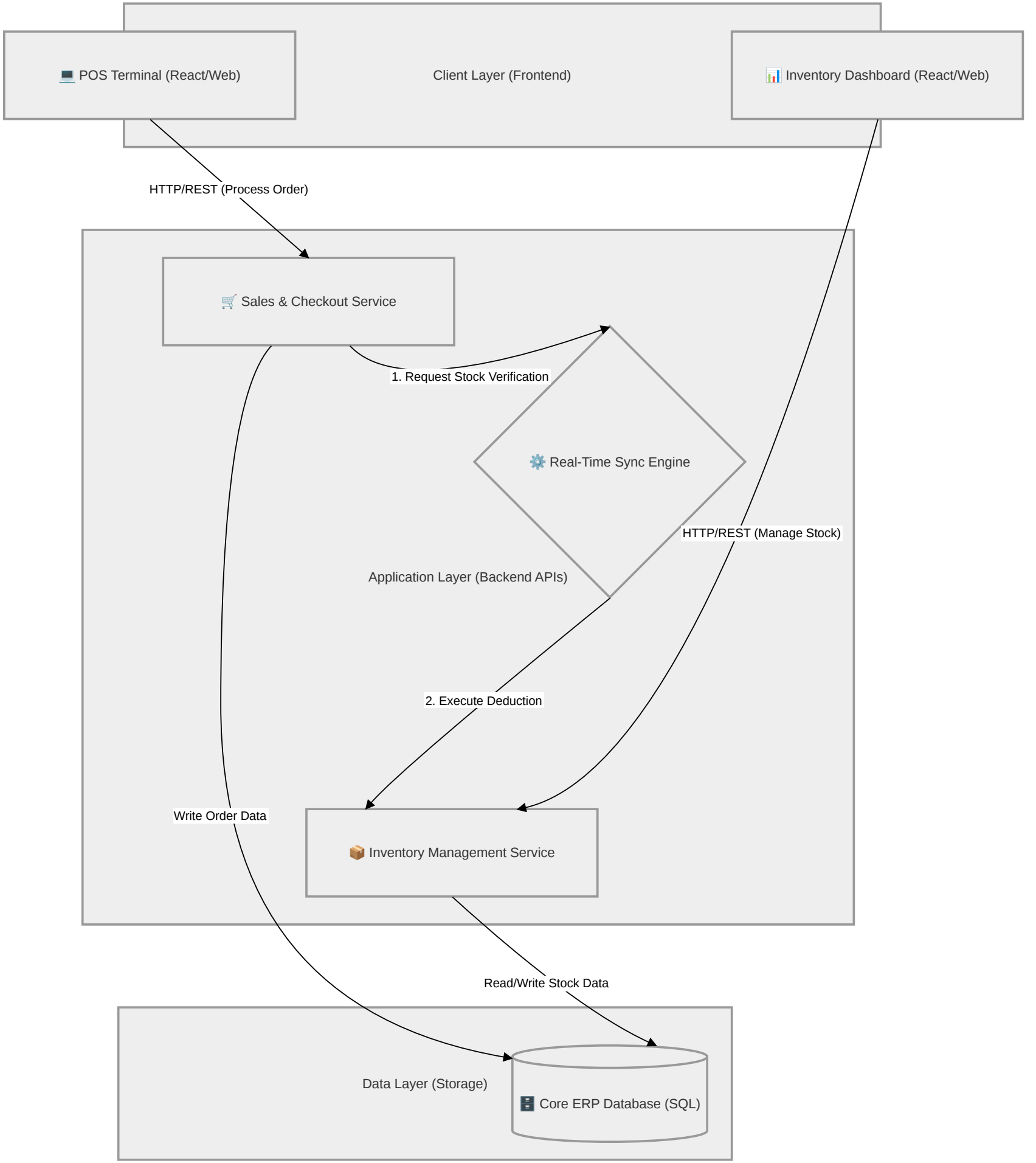
ERD

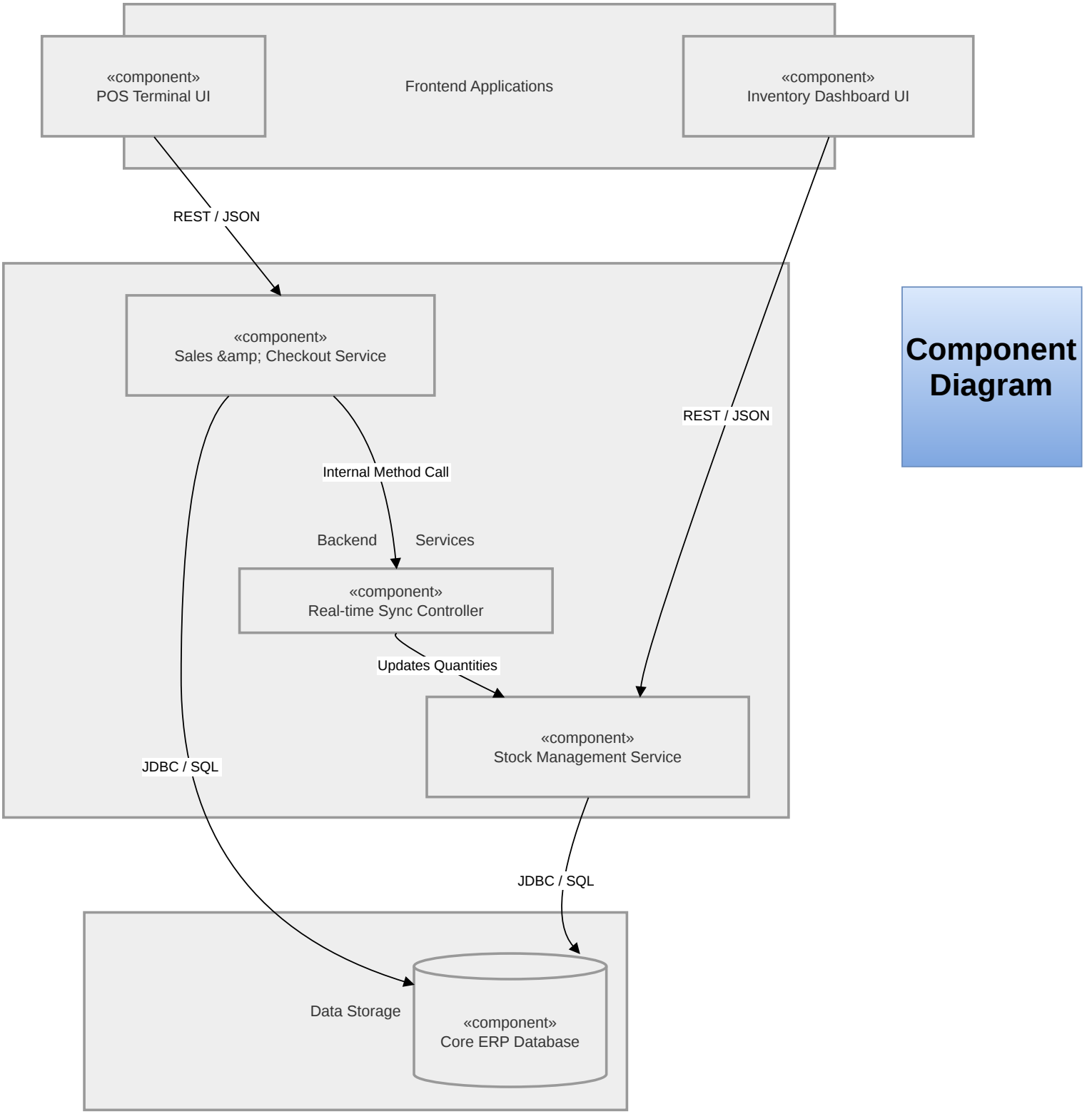

PRODUCT
int product_id PK
string name
float price
int current_stock
SALES_ORDER
int order_id PK
datetime transaction_date
float total_amount
string status
appears in
has history in
contains
triggers (Automated Deduction)
ORDER_ITEM
int item_id PK
int order_id FK
int product_id FK
int quantity
float subtotal_price
STOCK_TRANSACTION
int transaction_id PK
int product_id FK
int order_id FK
string transaction_type
int quantity_changed
datetime timestamp

Activity Diagram

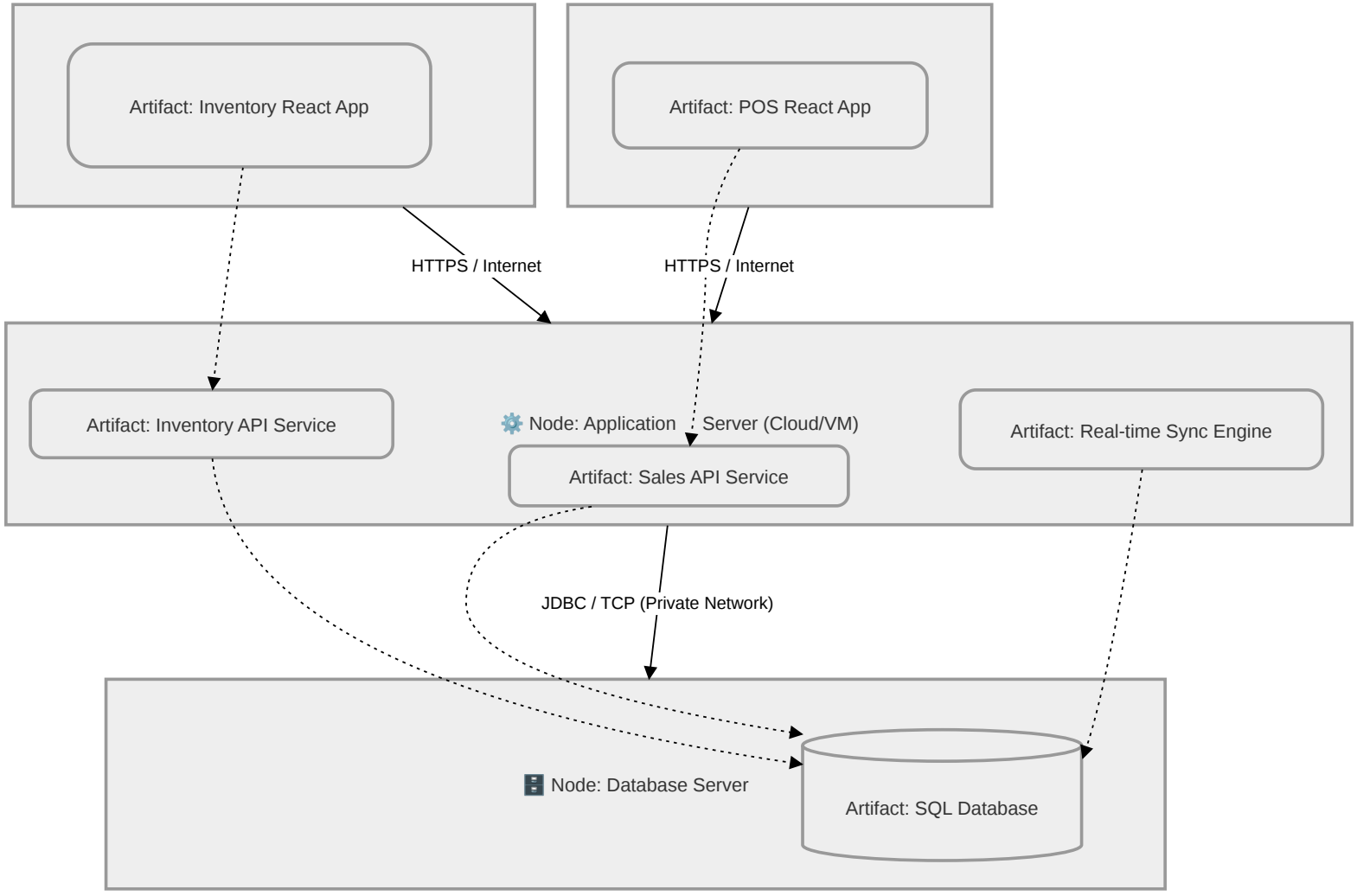


System Architecture Diagram

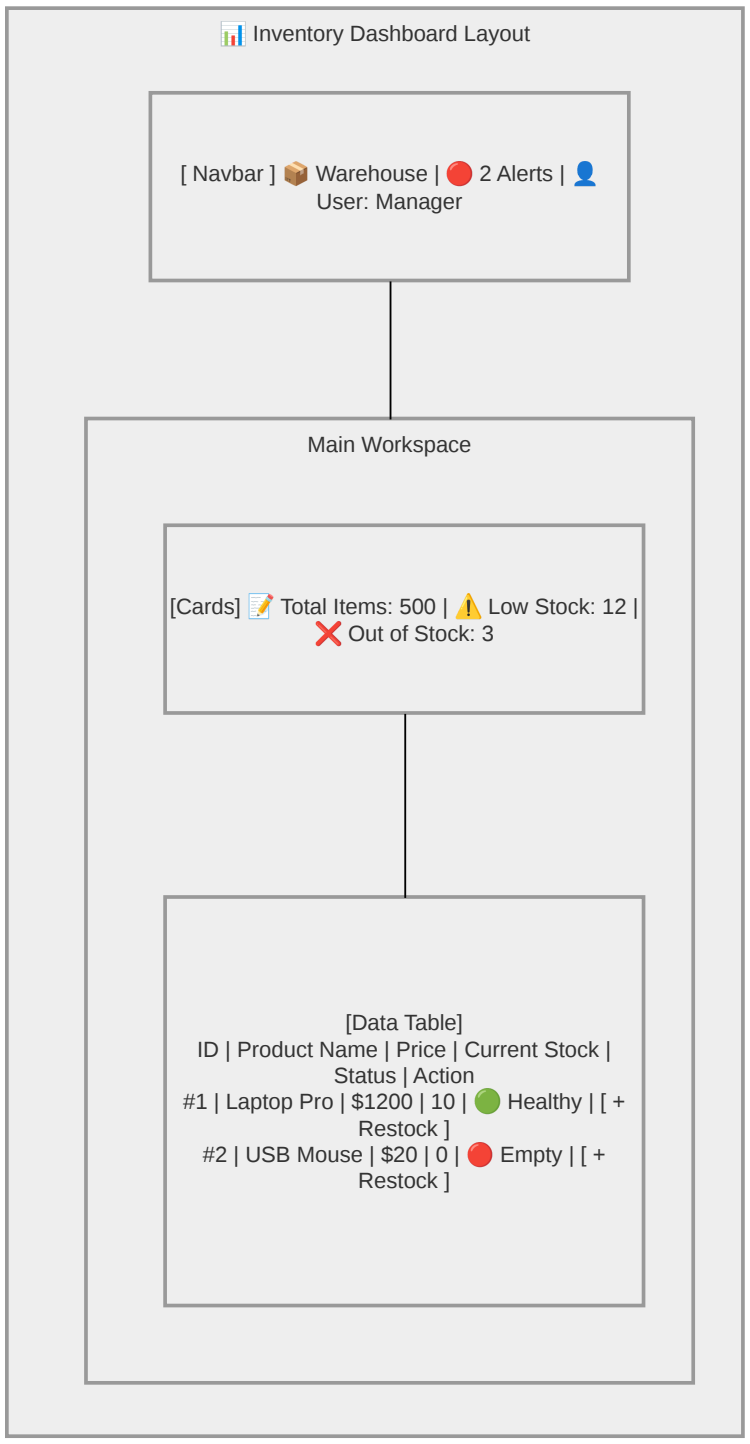


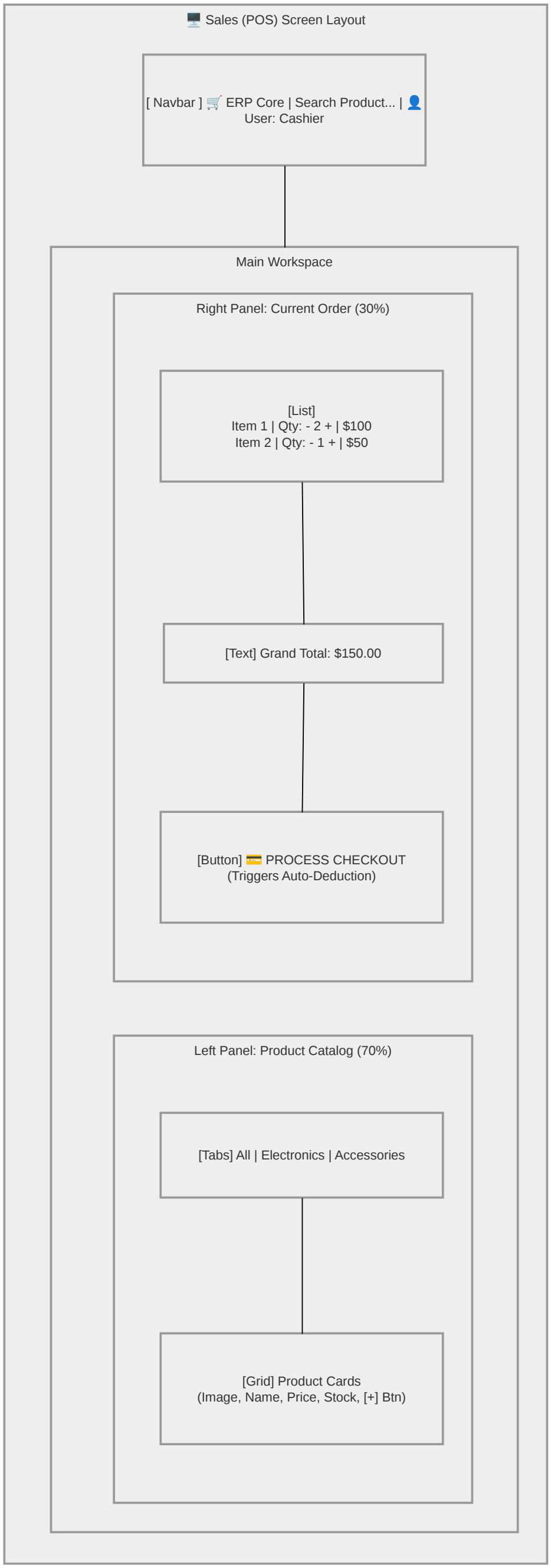

Frontend Applications
«component»
POS Terminal UI
«component»
Inventory Dashboard UI
REST / JSON
«component»
Sales & Checkout Service
REST / JSON
Internal Method Call
Backend
Services
«component»
Real-time Sync Controller
Updates Quantities
«component»
Stock Management Service
JDBC / SQL
JDBC / SQL
Data Storage
«component»
Core ERP Database
Component Diagram

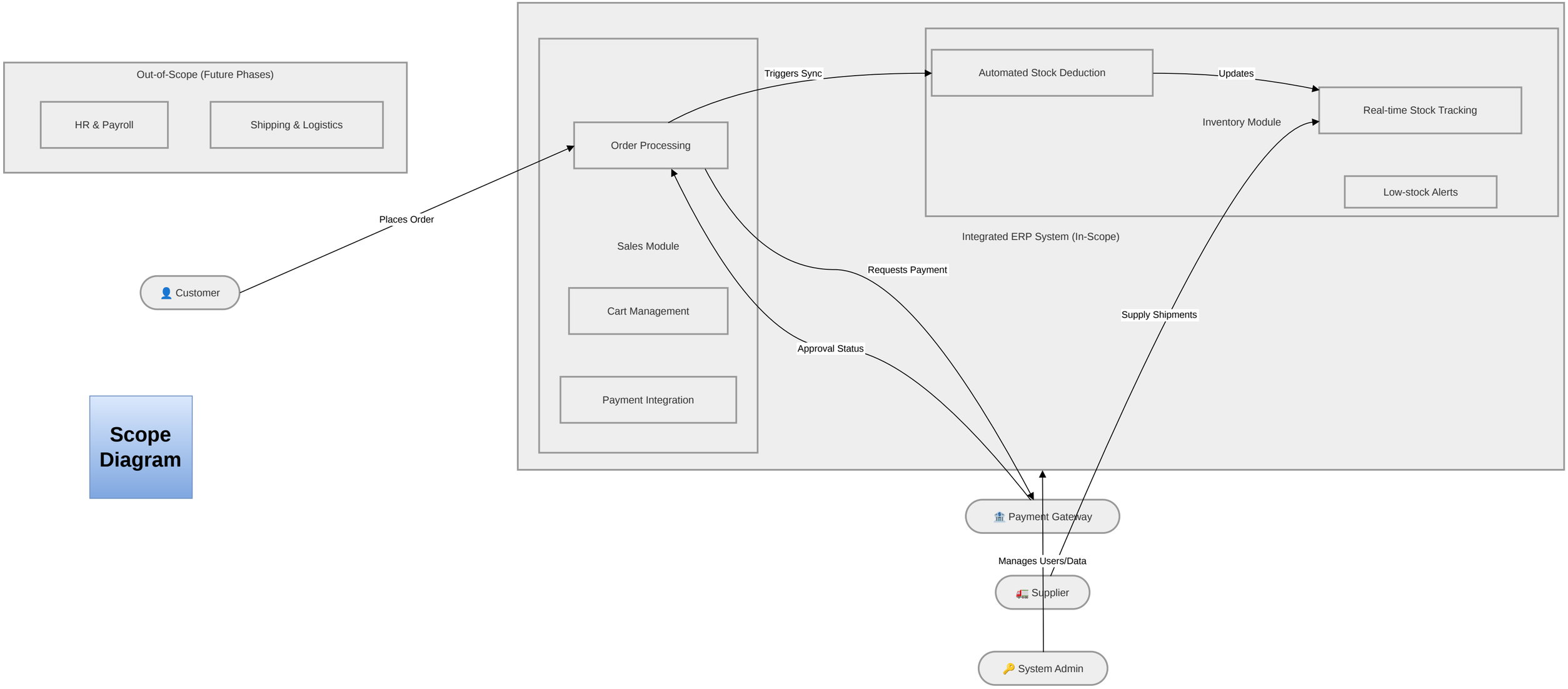
Deployment Diagram



UI Wireframes






Out-of-Scope (Future Phases)
HR & Payroll
Shipping & Logistics
Places Order
Customer
Triggers Sync
Order Processing
Sales Module
Cart Management
Payment Integration
Automated Stock Deduction
Updates
Real-time Stock Tracking
Inventory Module
Low-stock Alerts
Integrated ERP System (In-Scope)
Requests Payment
Approval Status
Supply Shipments
Payment Gateway
Manages Users/Data
Supplier
System Admin
Scope Diagram